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Research Paper

Assessment of Site Response and Building Damage Following the Mw 6.4 Copiapó Earthquake

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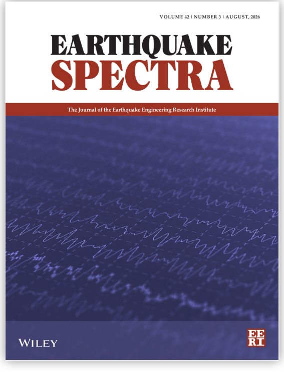

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References Related Information

Abstract

On June 6, 2025, an inslab earthquake of Mw 6.4 occurred in the north-central region of Chile, near the city of Copiapó, within the Atacama seismic gap. Despite its moderate magnitude, the event produced strong ground motions with recorded accelerations exceeding 0.4 g and widespread nonstructural damage. Due to these unusual consequences, we conducted a multidisciplinary field study, collecting in situ data, which provides an opportunity to characterize both local site conditions and structural behavior in a mature seismic gap region. Shallow shear-wave velocity profiles and predominant site periods were derived from ambient-vibration measurements to evaluate site amplification effects in Copiapó. Damage surveys revealed widespread nonstructural damage in designated evacuation zones, significantly impacting the functionality of numerous facilities. Finally, using ambient-vibration measurements, we estimated the structural properties of nine buildings (five damaged and four undamaged) of a residential housing complex located within the affected area. Despite their similar architecture and design, we identified systematic differences in the dynamic properties of the damaged buildings compared to the undamaged ones. On average, damaged buildings presented approximately 20% larger fundamental vibration periods and 10% higher damping ratios than undamaged buildings. The findings of this field survey provide insight into the vulnerability of buildings under moderate seismic events, whose seismic demand remains below that expected during a major earthquake.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Strong ground motions were obtained from the National Seismological Center of Chile (Centro Sismológico Nacional [CSN]) via their open-access website (https://evtdb.csn.uchile.cl/, last accessed November 2025). Seismic data processing and calculations were performed using MATLAB R2024b (**MATLAB 2024**). The map visualizations in Figure 1 benefited from the Generic Mapping Tools (**Wessel et al. 2019**) implemented in the PyGMT library (**Uieda et al. 2021**). The raw data and metadata for the ambient-vibration measurements used in this work are available at https://doi.org/10.5281/zenodo.18233849 (**Montalva et al. 2026**).

Supporting Information

References

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